Servo motor

DO | RASPBERRY PI | OUTPUT | MOVEMENT

What does it do?

A positional servo moves its arm, called a horn, to a chosen position and holds it there.

Short pulses on its signal wire set the position. The motor and position control are built into the case.

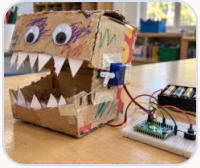
Use it for steering, pointing or opening a flap. A continuous-rotation servo controls speed instead; it cannot use this example to choose an angle.

Uses



Radio-controlled steering

DETECT Read steering command → **DECIDE** Which way should it turn? → **DO** Set the steering position



Cardboard chomping monster

DETECT Read feeding button → **DECIDE** Is the monster hungry? → **DO** Open its cardboard jaw



Horn	Signal	Power	Ground
Attach a light moving part	Usually orange, yellow or white	Usually red; about 5 V	Brown or black; join grounds

Raspberry Pi Foundation

Code

Raspberry Pi

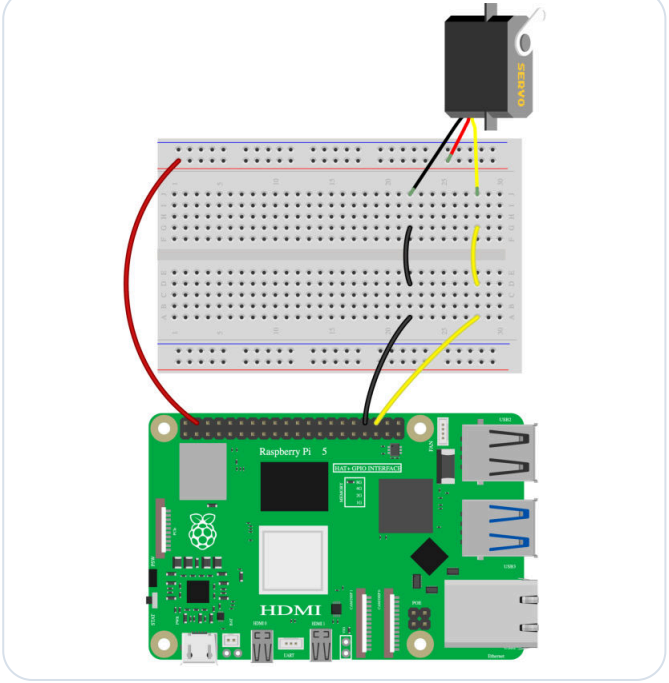
```
from gpiozero import Servo
from time import sleep

servo = Servo(16)
servo.mid()
sleep(2)
```

Test unloaded: the horn centres. Try min() or max() instead of mid(). Travel varies; stop if it binds or buzzes. Check lead order.

Wiring

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GPIO16 (pin 36) → signal; **5V** (pin 4) → top + rail → servo red; **GND** (pin 34) → servo −. Small **SG90-size** servo only: the Pi powers it.